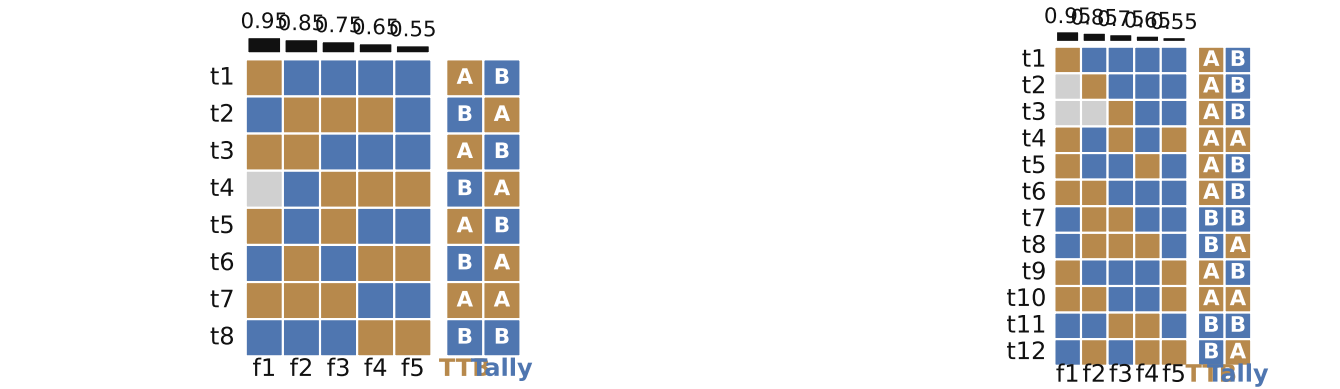


Case study: how the LLM narrows in on WADD
binary task · ground truth = WADD · noise = 0.0 · run3

Stage 1 — Experiment design
Exp 1 · proposed by TTB · 5 feats × 8 trials
Exp 2 · proposed by Tallying · 5 feats × 12 trials



Stage 2 — Metric proposal

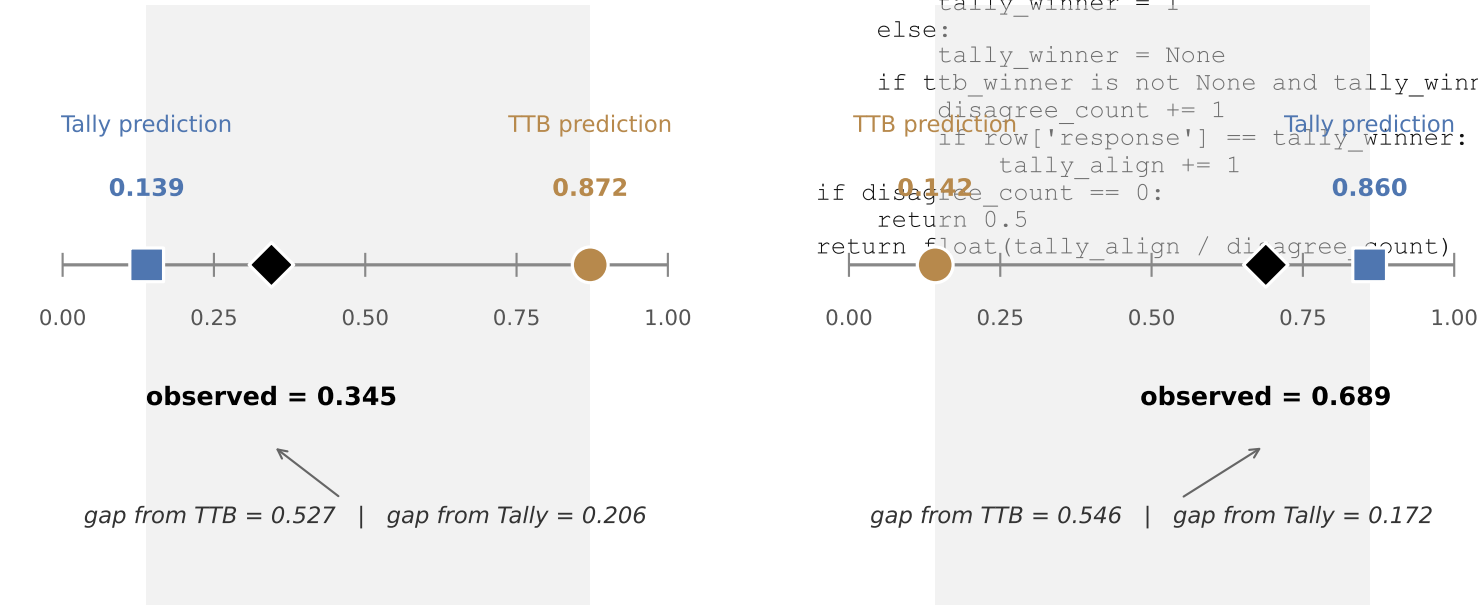
Exp 1 metric
Metric: fraction matching TTB on TTB-vs-Tallying disagreement trials.

Exp 2 metric
Metric: fraction matching Tallying on those disagreement trials.

```
A = np.array(data['option_a_ratings'].tolist())
B = np.array(data['option_b_ratings'].tolist())
diff = A - B
is_diff = diff != 0
has_diff = is_diff.any(axis=1)
first_diff_idx = np.argmax(is_diff, axis=1)
ttb_choice = np.where(diff[np.arange(len(diff)),
a_wins = np.sum(diff == 1, axis=1)
b_wins = np.sum(diff == -1, axis=1)
tally_choice = np.full(len(data), -1)
tally_choice[b_wins > a_wins] = 1
tally_choice[a_wins > b_wins] = 0
disagree = (has_diff) & (tally_choice != -1) & (ttb_choice != -1)
if np.sum(disagree) == 0:
    return 0.5
responses = data['response'].values
match = (responses[disagree] == ttb_choice[disagree])
return float(np.mean(match))

tally_align = 0
disagree_count = 0
for _, row in data.iterrows():
    a = np.array(row['option_a_ratings'])
    b = np.array(row['option_b_ratings'])
    a_tally = np.sum(a > b)
    b_tally = np.sum(b > a)
    ttb_winner = None
    for i in range(len(a)):
        if a[i] > b[i]:
            ttb_winner = 1
            break
        elif b[i] > a[i]:
            ttb_winner = 0
            break
        elif a[i] == b[i]:
            ttb_winner = None
            continue
    if ttb_winner is not None:
        if a_tally > b_tally:
            tally_winner = 1
        elif b_tally > a_tally:
            tally_winner = 0
        else:
            tally_winner = None
    if ttb_winner is not None and tally_winner is not None and ttb_winner != tally_winner:
        disagree_count += 1
    if row['response'] == tally_winner:
        tally_align += 1
if disagree_count == 0:
    return 0.5
return float(tally_align / disagree_count)
```

Stage 3 — Predictions bracket the observation (the gap)
Exp 1 — TTB-match rate — TTB-match
Exp 2 — Tally-match rate — Tally-match



Stage 4 — Arbiter reasons over both observations

Across both experiments, the data falls squarely between the predictions of Take The Best (TTB) and Tallying, though leaning slightly closer to Tallying.

“Instead, subjects likely use a weighted compensatory strategy where both the number of cues and their specific validities are integrated.”

verdict: new_theory → pi_1

Stage 5 — Theory proposer translates the gap into WADD

Weighted Additive (WADD) Theory: Decision-makers evaluate options using a fully compensatory strategy.

$$S_A = \sum_i v_i \cdot x_{A,i} \quad S_B = \sum_i v_i \cdot x_{B,i} \quad P(A) \propto \exp(\beta S_A)$$

Following the critic's feedback, the WADD mechanism is preserved entirely, but the parameter ranges for decision noise have been adjusted. The empirical metrics lie closer to 0.5 than the previous model's predictions.

“validity-weighted sum of cue values → compensatory decision-making”